A-A-1484A
January 12, 1988
Superseding
A-A-1484
December 14, 1982

COMMERCIAL ITEM DESCRIPTION

SOCKET, SOCKET WRENCH, NONPOWER, 3/8 INCH DRIVE
12-POINT, REGULAR LENGTH

The General Service Administration has authorized the use of this Commercial Item Description in preference to Type II, Class 2, Style A of Federal Specification GGG-W-641.

This commercial item description covers socket wrench sockets used for tightening and loosening bolts and nuts.

Salient characteristics:

The sockets shall conform to the requirements of ANSI B107.1 class 1, style A, double hexagon opening, 3/8 inch drive, in the sizes specified below.

Wrench openings: The sockets shall be sized to pass the gaging requirements of ANSI B107.17M.

National Stock Number	Opening (inches)	Bolt Clearance Hole Diameter (minimum) (inches)
5120-00-288-8155	1/4	0.200
5120-00-232-5711	5/16	0.265
5120-00-277-4251	11/32	0.320
5120-00-227-6702	3/8	0.328
5120-00-227-6703	7/16	0.390
5120-00-237-0977	1/2	0.453
5120-00-227-6704	9/16	0.515
5120-00-237-4973	5/8	0.578
5120-00-232-5706	11/16	0.650
5120-00-227-6705	3/4	0.703
5120-00-235-5807	13/16	--
5120-00-242-3330	7/8	--

Internal surfaces: The internal surfaces of the sockets shall be finished in accordance with the manufacturer's standard commercial practice.

Female socket drive end: The socket drive end shall be in accordance with ANSI B107.4, table 9, with the following exception: Delete note 1 of table 9, and substitute the following: "A concave recess on all four (4) sides of the female drive end shall be required. The recess shall be configured so as to mate with the spring loaded 0.188 inch diameter ball on the 3/8 inch male square drive. The recess shall have a minimum depth of 0.009 inch, and a minimum dimension of 0.079 inch for the lesser distance across the recess."

An optional recess type specified in figure 1 may be used in lieu of those figures and dimensions shown in table 9 of ANSI B107.4. This recess shall also be required on all four (4) sides of the female drive end.

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FSC 5120

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DRIVE SIZE		D_f		F_f (DIA)		A_f		C_f
INCH	MM	MAX	MIN	MAX	MIN	MAX	MIN	SEE TABLE 9. ANSI B107.4
3/8	10	.179	.162	.430	.420	.385	.378	

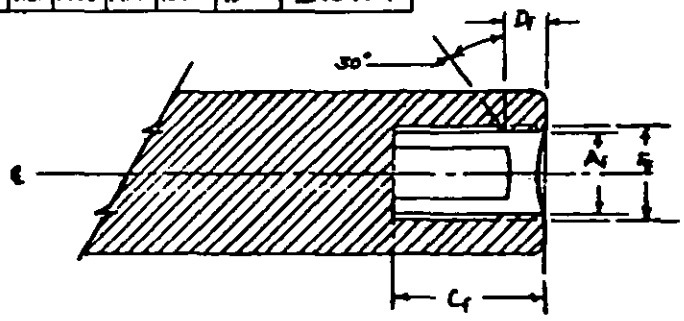


FIGURE 1. OPTIONAL RECESS TYPE

Note: F_f is applicable only to the optional recess type of female drive end.

Marking: In addition to any marking requirements specified, the manufacturer's name or identifying symbol, part number, and country of origin shall be etched or stamped in a permanent and legible manner on each socket.

Workmanship: The sockets shall be free from rust, fins, burrs, external sharp or rough edges, corners, and surfaces, and other defects which may impair their serviceability, durability, or appearance.

Quality Assurance Provisions.

Inspection. Unless otherwise specified in the contract or purchase order, the supplier shall be responsible for performing all inspection requirements. Sampling procedures shall be in accordance with MIL-STD-105, Sampling Procedures and tables for Inspection by Attributes, except where otherwise indicated herein. All sockets of one size offered for delivery at one time shall be considered a lot for the purpose of sampling. The sample unit shall be one socket.

Examination for visual requirements. The sockets shall be examined for conformance to design, workmanship and marking requirements with inspection level S-4 and an acceptable quality level (AQL) of 1.5 percent defective. Tears in the nut end or drive end caused by punching or broaching operations will be acceptable if the sockets meet the dimensional requirements when tested below.

Examination for dimensional requirements. The sockets shall be examined for the dimensional requirements by gaging as specified in ANSI B107.17M and B107.4 with inspection level S-3 and an AQL of 1.5 percent defective.

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Testing.

Hardness test. The sockets shall be tested for hardness as specified in section 5 of ANSI B107.1 with inspection level S-3 and an AQL of 1.5 percent defective.

Pull apart test. The sockets shall be tested to determine the force required to remove the sockets from a corresponding male drive tang in all four orientations. Any sockets requiring a removal force of less than four (4) pounds shall be considered defective. The inspection level shall be S-3 with an AQL of 1.5 percent defective.

Surface coating adhesion test. The sockets shall be examined for surface adhesion as specfied in paragraph 9, Heat Quench Test, of ASTM B571-84, Standard Test Methods for Adhesion of Metallic Coatings. The inspection level shall be S-3 with an AQL of 1.5 percent defective.

Load test. The sockets shall be load tested as specified in section 5 of ANSI B107.1 by applying a test load equal to 50 percent greater than the minimum test load specified in Table 2 thereof. Fragmentation of any socket into two or more separate pieces during the application of the test load shall be cause for rejection of the lot. Cracks appearing in any socket during application of any load between the minimum test load and the test load shall not be cause for rejection of the lot. The inspection level shall be S-2.

Regulatory requirement. In accordance with the section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers employees to undue hazards from the recovered materials.

Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be as specified in the contract or order.

Notes:

Except as otherwise noted herein, the issue of referenced documents in effect on the date of invitation for bid or request for proposals shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the contracting officer.

Military Standards are available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

Copies of ASTM test method are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

Copies of ANSI Standards are available from American National Institute, Inc., 1430 Broadway, New York, NY 10018.

MILITARY INTERESTS

Military Coordinating Activity:
Army-GL

Custodians:
Army-GL
Air Force-99

Review Activities:
Air Force-84
Army-AL

PREPARING ACTIVITY:

GSA-FSS

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